

Balraj Ashwath

◦ google.sites/balrajashwath ◦ balrajashwath.github.io
◦ linkedin/balrajashwath ◦ [github/BalrajAshwath](https://github.com/BalrajAshwath)

✉ balrajashwath98@gmail.com

☎ +91-9964952869

EDUCATION

- **National Institute of Technology Karnataka** Surathkal, IN
Bachelor of Technology in Electronics and Communications Engineering; GPA: 7/10 Aug. 2015 – Apr. 2019
- **S Cadambi Vidya Kendra** Bengaluru, IN
Indian Certificate of Secondary Education; Percentage: 92% Jul. 2010 – Apr. 2013

EXPERIENCE

- **Microland Limited** Bengaluru, IN
Machine Learning Engineer / Data Scientist Jul. 2019 - Present
 - **Error Log Pattern Analysis & Clustering:** Hierarchical clustering models were developed for unsupervised log pattern analysis using Ward's linkage criterion & cosine similarity features. Automated the cut-off selection procedure for choosing number of clusters from linkage matrix using Elbow method.
 - **Service-Ticket Topic Modeling & Error Log Classification:** LDA - an unsupervised topic model was developed for auto-tagging service tickets by identifying the top topics. Gensim, SpaCy & PyLDAvis packages were used for the same. Automated the selection of number of clusters using perplexity score. Supervised log classification models were implemented using Random Forests & CRF for classifying several standard log sources.
- **Video Analytics Lab, CDS Department, Indian Institute of Science** Bengaluru, IN
Research Intern (Mentors: Prof. Dr. R. Venkatesh Babu & K. Ram Prabhakar) Jul. 2019 – Present
 - **High-Resolution HDR Deghosting using CNNs (Paper submitted to CVPR 2020):** Proposed a novel deep neural network to produce high-quality ghost-free HDR for high-resolution images. The proposed method is fast, achieving 6 seconds to fuse a sequence of three images with 16MP resolution. Through experiments and ablations on different publicly available datasets, we show that the proposed method achieves state-of-the-art performance in terms of accuracy and speed.
 - **Domain Adaptation Techniques for HDR Image Deghosting:** Ongoing Research
- **Image Processing & Analysis Lab, National Institute of Technology Karnataka** Surathkal, IN
Undergraduate Researcher (Mentors: Prof. Dr. Shyam Lal & Karuna K. Eerapu) Jul. 2018 – Present
 - **Dense Refinement Residual Network for Road Extraction from Aerial Imagery Data:** Proposed a novel DRR-Net for semantic segmentation of roads. The proposed architecture effectively shares features across multiple scales and achieved a 2.74% increase in road accuracy with a ten-fold reduction in the number of parameters. Qualitative and quantitative results were reported on Massachusetts roads dataset.
 - **Dense CascadeNet for Road Extraction from Satellite Imagery:** Ongoing Research

PUBLICATIONS

- **Dense Refinement Residual Network for Road Extraction from Aerial Imagery Data:** *IEEE Access Journal*

PROJECTS & RELEVANT COURSES

Projects: Object Detection & Caption Generation, Face Recognition for Attendance Management, Human Pose Estimation, Visual Question Answering System, Human Activity Recognition, Tetris Game, Text Summarization with Attention, Image Style Transfer, Damage Identification on Post-Disaster Satellite Imagery, Artificial Image Colorizer, AI Flappy Bird (using NEAT), Tic-Tac-Toe AI (Minimax Game)

Relevant Courses: Stanford's CS231n - Convolutional Neural Networks, CS234 - Reinforcement Learning, CS224N - Natural Language Processing; MIT's 6.S191 - Intro to Deep Learning; Data Structures & Algorithms

PROGRAMMING SKILLS

- **ML Frameworks:** TensorFlow, PyTorch, Keras **Languages:** Python, Java, C, C++, JavaScript, Go

HONORS & AWARDS

- Intel Edge AI Scholarship '19, Merit-cum-Means Scholarship, FFE Scholar, HackWithInfy '18 Finalist